

Arguments must move at least once: Evidence from Split Constructions

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1. Introduction

This paper argues that arguments never remain in the position where they are base generated but are obligatorily moved at least once (Kayne 2010), as summarized in (1):

- (1) **The Obligatory Movement Hypothesis:** All arguments undergo at least one instance of syntactic movement.

I provide novel evidence for this hypothesis from the domain of discontinuous PPs, hereafter ‘P-splits’, in Modern Greek. While Angelopoulos (2019), Angelopoulos (2025), Angelopoulos & Michelioudakis (2023), Angelopoulos & Michelioudakis (2025) explored these constructions mostly with intransitive predicates, this paper expands the empirical scope to transitive predicates. The extended analysis reveals that the mechanics of P-split formation are incompatible with the assumption that direct objects can remain in situ. Instead, the data strongly suggest that objects must move, and this movement has predictable consequences for how to analyze certain orders in Greek, e.g., VSO.

The paper is structured as follows. Section 2 provides background on the formation and analysis of P-splits in Greek, establishing a remnant movement account based on intransitive constructions. Section 3 presents new data from transitive predicates. These sections show how intervention effects in P-split formation reveal the obligatory movement of objects and support the hypothesis in (1). Section 4 discusses the broader consequences of this analysis for the syntax of Greek, particularly concerning the position of post-verbal subjects. Section 5 is a short note on how P-splits can be formed in Greek and section 6 concludes.

2. Background: P-Splits as Remnant Movement

In previous work (Angelopoulos 2019, Angelopoulos 2025, Angelopoulos & Michelioudakis 2023, Angelopoulos & Michelioudakis 2025), we examined discontinuous constituents arising from Greek possessor constructions. A genitive possessor typically follows the noun it modifies, as in (2a), but can appear prenominal under specific discourse conditions like focus or topic, (2b). The possessor can also be separated from the possessum, as shown in (2c), where *tu Kusturica* ‘Kusturica’s’ is separated from the accusative object *tin tenia* ‘the film.’

- (2) a. Idhe tin tenia tu Kusturica.
saw.3SG the.ACC film.ACC the.GEN Kusturica.GEN
‘He saw the film of Kusturica.’
b. Idhe tu Kusturica tin tenia.
saw.3SG the.GEN Kusturica.GEN the.ACC film.ACC
‘He saw KUSTURICA’s film.’
c. Tu Kusturica idhe tin tenia.
the.GEN Kusturica.GEN saw.3SG the.ACC film.ACC
‘He saw KUSTURICA’s film.’

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A more complex discontinuity, which I term a ‘P-split’, emerges when these possessor constructions are embedded within a PP, as in (3a). The preposition can be fronted along with a *wh*-possessor, stranding the possessed noun phrase (the ‘possessum’) post-verbally, as shown in (3b). Likewise, the possessor can be fronted together with the P to express focus, (3c).

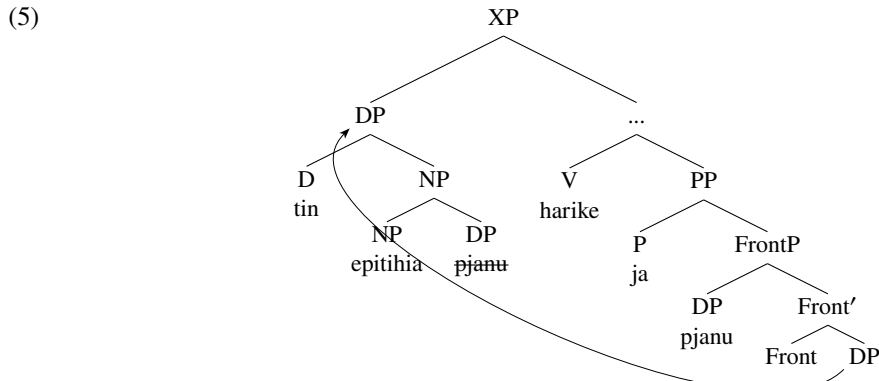
- (3) a. Xarike ja tin epitixia tu Kusturica.
was-happy.3SG for the.ACC success.ACC the.GEN Kusturica.GEN
‘He was happy for Kusturica’s success.’
- b. Ja pjanu xarike tin epitixia?
for whose.GEN was-happy.3SG the.ACC success.ACC
‘For whose success was he happy?’
- c. Ja tu Kusturica xarike tin epitixia.
for tu.GEN Kusturica.GEN was-happy.3SG the.ACC success.ACC
‘For KUSTURICA’s success was he happy?’

Following Angelopoulos (2019), Angelopoulos (2025), Angelopoulos & Michelioudakis (2023), Angelopoulos & Michelioudakis (2025), I analyze P-splits as instances of remnant movement, involving three independently motivated steps:

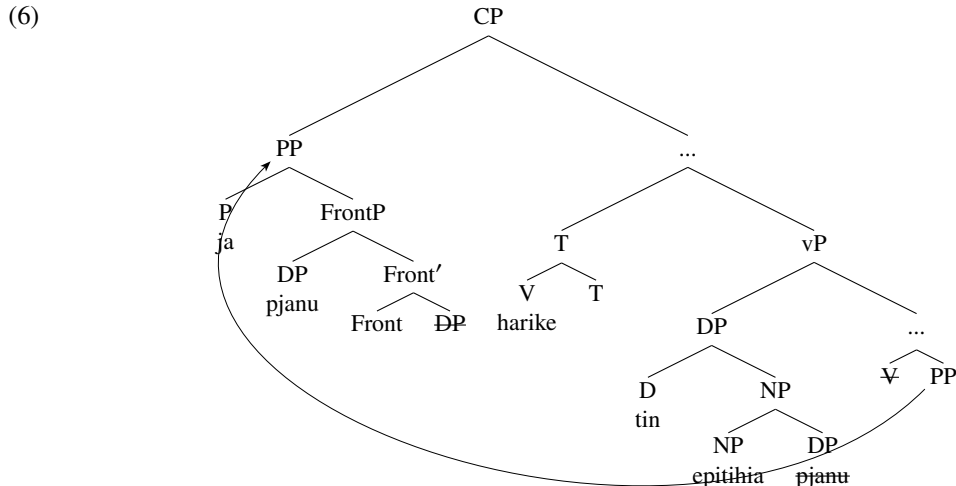
Step 1: Possessor Extraction: The genitive possessor moves to a DP-peripheral position, let us call it FrontP. This is the same movement responsible for the prenominal possessor order seen in (2b).

- (4) [_{FrontP} pjanu_j [_{Front'} Front [_{DP} tin_D epitixia_{NP} t_j]]]

Step 2: Remnant DP Movement: The larger DP, now a remnant, undergoes A-movement to a functional projection in the clause’s middle-field. Let us call it Spec,XP for convenience.



Step 3: Remnant PP Fronting: The entire PP, which now contains the trace of the remnant DP, undergoes A-bar movement to the left periphery of the clause (e.g., Spec,CP).



The proposed multi-step derivation is supported by direct evidence from the behavior of different types of genitives. The analysis predicts that only genitives capable of undergoing the initial DP-internal fronting (Step 1) should be able to form a P-split.

This prediction is borne out. Consider non-possessive, appositive genitives like *ton Vrixelon* in (7a). Unlike true possessors, these genitives cannot be moved to a pre-nominal position, as shown by the ungrammaticality of (7b) (Angelopoulos & Michelioudakis 2025).

- (7) a. Efije apo tin poli ton Vrixelon.
left.3SG from the.ACC city.ACC the.GEN Brussels.GEN
‘He left from the city of Brussels.’
b. *Efije apo ton Vrixelon tin poli.
left.3SG from the.GEN Brussels.GEN the.ACC city.ACC
‘He left from the city of Brussels.’

Because this initial DP-internal fronting step (Step 1) is a necessary precondition for creating the remnant DP, the analysis correctly predicts that such constructions cannot form a P-split. The ungrammaticality of (8) confirms this.

- (8) *Apo ton Vrixelon efije tin poli.
from the.GEN Brussels.GEN left.3SG the.ACC city.ACC
‘From city of Brussels, he left.’

In short, the inability of appositive genitives to undergo DP-internal fronting directly correlates with their inability to participate in P-splits, strongly supporting the claim that the former is a prerequisite for the latter.

This analysis has so far been based on intransitive verbs like *xarike* ‘was happy.’ The central goal of this paper is to investigate whether P-splits are possible with transitive predicates, a question that has not yet been explored. As we will see, the constraints on their formation in such environments provide strong evidence for the Obligatory Movement Hypothesis in (1).

3. P-Splits with Transitive Predicates

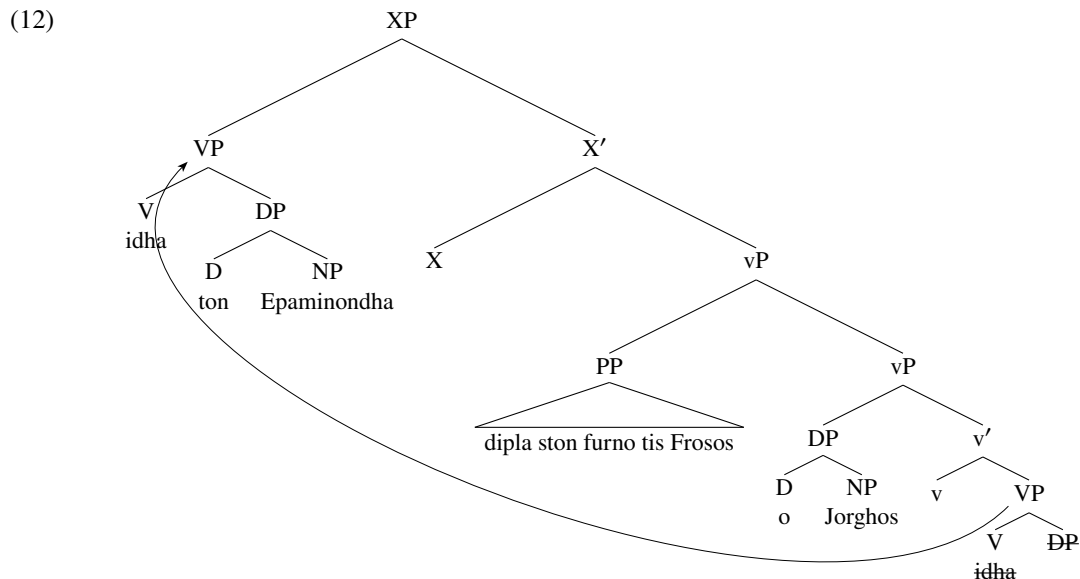
We turn to transitive predicates. Consider a baseline sentence where the verb takes a direct object and a locative PP modifier (9a). As (9c) and (9b) show, forming a P-split from the locative PP is strictly ungrammatical when a definite direct object is present. Note that the same effect arises with all different kinds of PPs that can form a P-split; the split is not allowed when the PP is combined with a transitive predicate.

- (9) a. Idha ton Epaminondha dhipla s-ton furno tis Frosos.
saw.1SG the.ACC Epaminondhas.ACC next to-the.ACC bakery.ACC the.GEN Frosos.GEN
‘I saw Epaminondhas next to Froso’s bakery.’
b. *Dhipla se pjanu idhes ton Epaminondha ton furno?
next to whose.GEN saw.2SG the.ACC Epaminondhas.ACC the.ACC bakery.ACC
‘Next to whose bakery did you see Epaminondhas?’
c. *Dhipla se pjanu idhes ton furno ton Epaminondha?
next to whose.GEN saw.2SG the.ACC bakery.ACC the.ACC Epaminondhas.ACC
‘Next to whose bakery did you see Epaminondhas?’
d. Dhipla se pjanu ton idhes ton furno?
next to whose.GEN 3SG.ACC.M saw.2SG the.ACC bakery.ACC
‘Next to whose bakery did you see him?’

This blocking effect is not a surface constraint against two adjacent accusative definite DPs, e.g., *ton furno* and *ton Epaminondha* in (9c) and (9b), with identical features. The construction remains ungrammatical if the direct object is indefinite (10), and it becomes grammatical in contexts where two definite DPs with identical features are adjacent, as long as one is a temporal adverbial (11), i.e. *cthes to vradhi* ‘last night’.

- (10) a. Idha ena pedhi dipla s-ton furno tis Frosos.
saw.1SG a kid.ACC next to-the bakery.ACC the.GEN FROSOS.GEN
'I saw a girl next to Frosos' bakery.'
- b. *Dhipla se pjanu idhes ton furno ena pedhi?
next to whose.GEN saw.2SG the.ACC bakery.ACC a.ACC child.ACC
'Next to whose bakery did you see a child?'
- c. *Dhipla se pjanu idhes ena pedhi ton furno?
next to whose.GEN saw.2SG a.ACC child.ACC the.ACC bakery.ACC
'Next to whose bakery did you see a child?'
- (11) a. Ekatsa chthes to vradhi dipla sto aftokinito tis Elenis.
sat.1SG yesterday the night.ACC next to-the car.ACC the Eleni.GEN
'I saw yesterday night next to Eleni's car.'
- b. Dipla se pjanu ekatses cthes to vradhi to aftokinito?
next to whose.GEN sat.2SG yesterday the night.ACC the car.ACC
'Next to whose car did you sit last night?'

The pattern becomes clear under the Obligatory Movement Hypothesis. I propose that direct objects must always move from their base position to a licensing position in the middle-field—position targeted by the remnant DP in P-split formation (Step 2). Let us consider the derivation underlying the baseline sentence in (9a). The object is always moved to the middle-field, not on its own, but smuggled inside the VP where it is base generated, as shown in (12) (see discussion in Section 4 on why object movement takes place inside the VP).



To form a P-split, Step 1-3 must take place. Step 2 requires movement of the accusative DP, *ton furno* 'the bakery,' from inside the PP to the middle-field. Given this, the ungrammaticality of (9b) is straightforward: the direct object *ton Epaminondha* 'Epaminondha' has moved inside the VP to the middle-field first, but since there is only one position for accusative objects, movement of *ton Epaminondha* there blocks movement of *ton furno* from inside the PP (Step 2) to the same position. Similarly, (9c) is ungrammatical because *ton furno* has already moved from the PP into the middle-field, as indicated by its position preceding the direct object *ton Epaminondha*. This movement in turn prevents the direct object from raising to the same position. The question that arises is why cliticization of the direct object suddenly allows the formation of a P-split, (9d). As in Angelopoulos & Sportiche (2021), I assume that clitics are heads situated above T. In simple cliticization, they attract a silent pronoun to the specifier of this clitic head from the argument position of the verb. Crucially, this pronoun does not move

through Spec,XP, so formation of the P-split becomes possible with cliticized direct objects because this middle-field position is available for movement of *ton furno* inside the PP to this position.

The fact that P-splits are impossible in (9b), (9c) and (10) constitutes the core argument of this paper. If the direct object could optionally remain in situ, the middle-field position should be available for movement to the middle-field, as required by P-split formation, and the derivation should converge. The fact that it systematically fails demonstrates that the object must move to this middle-field position.

The grammaticality of (11) is also explained. High temporal adverbs like *xthes to vradhi* ‘last night’ are merged higher than the middle-field position targeted by argument movement. As such, they do not occupy the middle-field position required for the formation of a P-split. This correctly predicts that temporal PPs, which are also merged high, cannot themselves undergo P-splitting, as they cannot undergo the crucial Step 2 movement to the middle-field—a fact noted in Angelopoulos (2019, 2025).

4. Discussion: Consequences for VSO orders

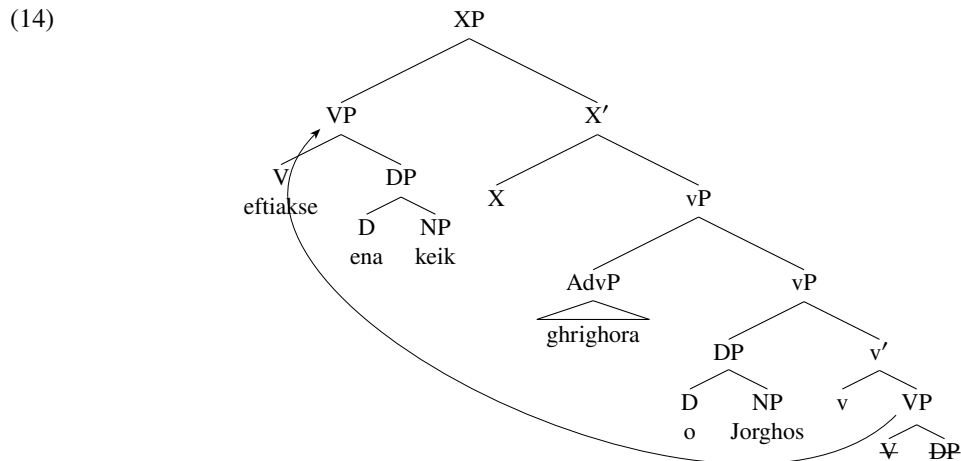
The conclusion that arguments obligatorily move to a middle-field licensing position has significant consequences for the analysis of Greek syntax. The canonical assumption is that objects can remain in their V-complement position and subjects can remain in Spec,vP. This view is central to the standard derivation of VSO word order in null subject languages: V moves to T, stranding the subject in Spec,vP and the object in its base position (Alexiadou & Anagnostopoulou 2001, see also Roussou & Tsimpili 2006).

Our findings demonstrate this analysis is untenable. The object must move out of the VP to the middle-field. Consequently, in a VSO sequence, the subject cannot be in Spec,vP, as the object has already moved past it. The subject must also have moved to a position higher than the object’s landing site.

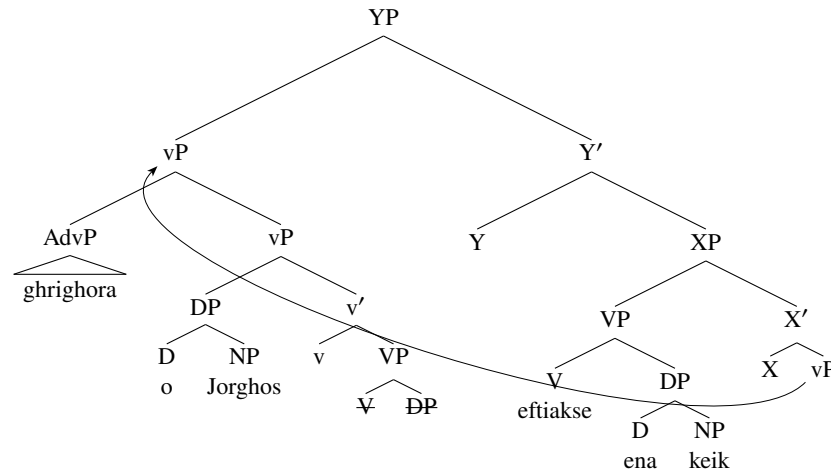
One common argument for a low, in situ subject in VSO orders is its ability to follow vP-level adverbs like manner adverbials:

- (13) Eftiaxe ghrigora o Yorgos ena keik.
made.3SG quickly the.NOM Giorgos.NOM a.ACC cake.ACC
‘Giorgos quickly made a cake.’

However, this order is compatible with our analysis. The object *ena keik* ‘a cake’ is moved inside the VP to the middle-field, higher than the vP, and higher than the attachment position of the manner adverb, as shown in (14). Subsequently, the subject can pied-pipe the vP containing the manner adverb and the object trace to a position higher than the object, as shown in (15). V is subsequently moved to T through Y, not shown here, deriving the observed V-Adv-S-O order.



(15)



Pied-piping is only optional so the adverb can be stranded in a low position. Consequently, the order V-S-O-Adv is also possible with a neutral intonation:

- (16) Eftiaxe o Yorgos ena keik ghrigora.
 made.3SG the.NOM Giorgos.NOM a.ACC cake.ACC quickly
 'Giorgos quickly made a cake.'

It is not entirely clear though how this order is derived under an analysis in which postverbal subjects only occur in Spec,vP and objects are in their base generated position.

A final point regarding the locality of object movement in Greek is needed. We have operated under the assumption that the object's movement to the middle-field occurs inside the VP. This assumption is motivated by locality considerations: movement of the direct object across the subject would constitute a minimality violation. This issue is circumvented if the object is moved within the VP, as we have assumed. Furthermore, there is no evidence that the object moves on its own to a position higher than the subject at any stage of the derivation. If such movement were possible, we would expect it to obviate WCO effects. However, WCO violations persist across SVO, VSO, and even VOS orders, as shown in (17). The persistence of WCO effects strongly suggests that the object never occupies a c-commanding position over the subject, confirming that its movement is always contained within the VP.

- (17) a. I tenia tu_{i/j} tapinose kathe skinotheti*_i.
 the movie.NOM his.GEN humiliated.3SG every director.ACC
 'His movie humiliated every director.'
 b. Tapinose i tenia tu_{i/j} kathe skinotheti*_i.
 humiliated.3SG the movie.NOM his.GEN every director.ACC
 'His movie humiliated every director.'
 c. Tapinose kathe skinotheti*_i i tenia tu_{i/j}.
 humiliated.3SG every director.ACC the movie.NOM his.GEN
 'His movie humiliated every director.'

5. Discontinuous phenomena (a final note)

Let us reconsider the status of example (9c), repeated here as (18a). While ungrammatical with a neutral intonation, the sentence's acceptability improves markedly when a sharp prosodic break separates the stranded nominal *ton furno* 'the bakery' from the direct object *ton Epaminondha* 'Epaminondha,' (18b).

- (18) a. *Dhipla se pjanu idhes ton furno ton Epaminondha?
 next to whose.GEN saw.2SG the.ACC bakery.ACC the.ACC Epaminondhas.ACC
 'Next to whose bakery did you see Epaminondhas?'
 b. Dhipla se pjanu idhes ton furno ton Epaminondha?
 next to whose.GEN saw.2SG the.ACC bakery.ACC the.ACC Epaminondhas.ACC
 'Next to whose bakery did you see Epaminondhas?'

- b. Dhipla se pjanu idhes ton furno, ton Epaminondha?
 next to whose.GEN saw.2SG the.ACC bakery.ACC the.ACC Epaminondhas.ACC
 ‘Next to whose bakery did you see Epaminondhas?’

This prosodic effect is characteristic of extraposition. I tentatively propose that the amelioration observed in (18b) stems from the fact that extraposition involves movement of the extraposed element to a position higher than the middle-field. The derivation proceeds as follows: the direct object *ton Epaminondha* undergoes movement (inside the VP) to the left periphery of the clause. This could either be rightward movement or movement to a specifier in the left periphery. In the latter case, the object, *ton Epaminondha*, occurs as the last element in a clause as a result of remnant movement of the rest of the clause around *ton Epaminondha*. In both cases the result is the same: since *ton Epaminondha* moves to the left periphery, the middle-field position is empty thus allowing the formation of a P-split. It is an open question why the direct object in this case does not move through the intermediate position. Nevertheless, suffice it to say for now that arguments can also be licensed in the left periphery, e.g. in cases where they are base generated there, as is the case with Hanging Topics.

6. Conclusion

This paper has argued for the Obligatory Movement Hypothesis, according to which all arguments must undergo at least one movement step in the course of the derivation. Drawing on novel evidence from P-splits in Modern Greek, I have shown that the formation of discontinuous PP constructions crucially depends on prior movement of nominal arguments to the middle-field. The ungrammaticality of P-splits in the presence of direct objects, and the rescue effects created by cliticization or extraposition, strongly support the claim that arguments cannot remain in their base-generated positions.

The consequences of this conclusion extend beyond the narrow domain of P-splits. If arguments obligatorily move, then the traditional view of Greek VSO orders as reflecting subjects and objects in situ must be revised. Instead, both objects and subjects occupy higher positions than previously assumed, and the observed word orders are derived by further remnant movement operations.

More broadly, the evidence from Greek supports the view that movement is not an optional mechanism available to satisfy information-structural requirements, but a necessary step in the licensing of arguments. The study of discontinuous phenomena thus continues to reveal fundamental properties of syntactic architecture: movement is not the marked option, but the default path by which arguments are licensed in the syntactic derivation.

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